Evaluation of Zero-shot Text Classification Methods in Turkish

by

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Abstract

Effective learning schemes such as fine-tuning, zero-shot, and few-shot learning, have been widely used to obtain considerable performance with only a handful of annotated training data. In this paper, we present a unified benchmark for zero-shot text classification in Turkish, propose a zero-shot learning approach to facilitate the problem of text classification. Additionally, we show that use of monolinguals models improve zero-shot text classification performance over multilingual models that were used in previous studies of zero-shot Turkish text classification.

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List of Abbreviations

NLP	Natural language processing
NSP	Next sentence prediction
NLI	Natural language inference
MLM	Masked language modeling
ZSL	Zero-shot learning
ZS	Zero-shot
TC	Text classification

1 Introduction

Conventional text classification that plays important role in the field of language processing and text mining, classify documents into predetermined categories. Recently, the wide range of sources that generate huge amounts of data and the most of the remarkable benchmarks stand on a considerable amount of labeled data. Thus the requirement of automatic annotation has become increasingly important because manually annotating is time-consuming and expensive.

In recent years, different learning schemes such as fine-tuning, zero-shot, few-shot learning ([1, 2, 3, 4] have attracted considerable attention because of tremendous success in natural language understanding tasks beside machine learning, prominent one of the traditional methods, to address classification. Traditionally, Zero-Shot Learning (ZSL) is designed a model to learn a classifier on set of classes and predict unseen classes.

One of the primary constraints is the lack of a unified benchmark for zero-shot Turkish NLP, thus new methods cannot be easily compared and iteratively improved.

In this study, we evaluate three approaches of zero-shot classification, namely NLI, NSP and MLM, on 7 Turkish datasets from three main categories: topic, sentiment, and emotion. We use pre-trained Turkish and multilingual transformer models.

The main contributions of this paper are:

- A unified benchmark for zero-shot Turkish text classification task, which consists of 7 datasets from 3 categories: topic, emotion and sentiment.
- Another approach to ZS text classification, that utilizes masked language modeling (MLM) and pre-trained word embeddings.

The rest of this paper is organised as follows: Section 2 presents preliminaries on ZSL. The project design is presented in Section 3. Datasets and Methodology parts are proposed in Sections 4 and 5. Experimental Setup is explained in Section 6. Finally, results are of our experiments, conclusion and future research directions are given in Section 7 and 8 respectively.

We publicly share our code repository for implemented the methods and experiments[1], all datasets[2] used in the experiments, and a demo application[3].

[1] https://github.com/emrecncelik/zeroshot-turkish
[2] https://drive.google.com/drive/folders/1J_4dBQy1mQEbvhyHLfrlc6akujg9DEw3?usp=sharing
[3] https://huggingface.co/spaces/emrecan/zero-shot-turkish

2 Related Works

In recent years, dataless classification, recently known as zero-shot classification [5] is attractive and has shown promising results by enhancing state-of-the-art performance in classification problem. In study [6], the author use Explicit Semantic Analysis for the semantic representation of texts and Wikipedia as the source of world knowledge. Dataset is separated into multiple binary classification problems and results show 85.29% average accuracy on 20NG dataset [7] and 88.62% on Yahoo! Answers dataset, outperforming the supervised baseline model which was trained with 10 examples. Extended version to multi-class and multi-label settings is proposed by [8]. The authors utilize label hierarchies and compare different semantic representation methods like ESA and neural network word embeddings from [9] on the new task of dataless classification. The best performing method, ESA, is combined with a bootstrapping step and has 83.37% average F1 score on 20NG dataset.

[10] propose a zero-shot learning framework for the semantic utterance classification task. The authors train a neural network with Bing search query click logs and hypothesize that queries with similar meanings would correspond to same url clicks. Using the last hidden layer of the neural network for embeddings for semantic representation of texts and labels, the authors achieve higher scores than a supervised linear SVM model in case of less than 100 training examples.

[11], the authors present GPT-2 [12], a generative language model, to approach the zero-shot classification problem as an multiple choice question answering task. They collect Reddit post titles and outbound weblinks of the posts and use them to create a multiple choice question answering dataset such as SST-2, AGNews, DBPedia.

In study [13], the authors approach the zero-shot text classification problem as a textual entailment problem. The entailment method compared with other methods such as ESA, utilizing semantic similarity of labels and texts with Word2Vec, and fine-tuning BERT [14] on binary classification with a dataset collected from Wikipedia. Entailment approach achieved the highest accuracy scores on situation and emotion datasets, while having comparable but a lower score to binary BERT on the topic dataset.

[15] points out the issues in using the entailment approach for zero-shot classification. The authors show that models trained on the NLI task may have learnt patterns that are specific to NLI datasets which reduce the performance on zero-shot classification task. The authors also compare next sentence prediction (NSP) method, pre-training task of BERT, for classification with the entailment method and show that fine-tuning on NLI task

does not bring any performance gain on many datasets.

In yet another study [16], the authors dove deep into the method of using NSP for zero-shot NLP tasks. NSP method is applied on wide variety of tasks that can be broadly categorized as cloze-style, sentence pair and single sentence. Considering classification as a single sentence task, they outperform GPT-3 [1] and PET [3] in zero-shot setting for all single sentence tasks.

[17] used a version of XLM-RoBERTa [18], a multi-lingual language model fine-tuned on XNLI [19], a multi-lingual NLI dataset, to make predictions on Turkish classification datasets. Datasets they used for classification are TTC-3600 [20], a news dataset in Turkish, and TTD, a sentiment dataset which consists of Turkish tweets. Reported maximum accuracies of the entailment zero-shot method on these datasets are respectively 65.5% and 91.5%. The study also compared the zero-shot performances with the performances acquired by fine-tuning BERT on the classification datasets, they report that training on 1800 of training samples from TTC-3600 and 50 samples from TTD is enough to match the zero-shot performance.

[21] created Turkish versions of SNLI [22] and MultiNLI [23] datasets using machine translation, and assured that the translations are of high quality. The autors fine-tune and compare English, multi-lingual and Turkish versions of the BERT model on the datasets and conclude that pre-training on the specific target language improves performance on the downstream NLI task.

The above-mentioned studies based on zero-shot learning mostly used methods that require a training prior to inference. In this study, we present a different approach that does not require prior training on NLI, and we use monolingual models instead of the multilingual ones for Turkish.

3 Design

In this section, we show the steps we follow to apply different methods to zero-shot classification setting and our project timeline. Figure 1 shows the main processes we follow and Figure 2 shows our planned project timeline.

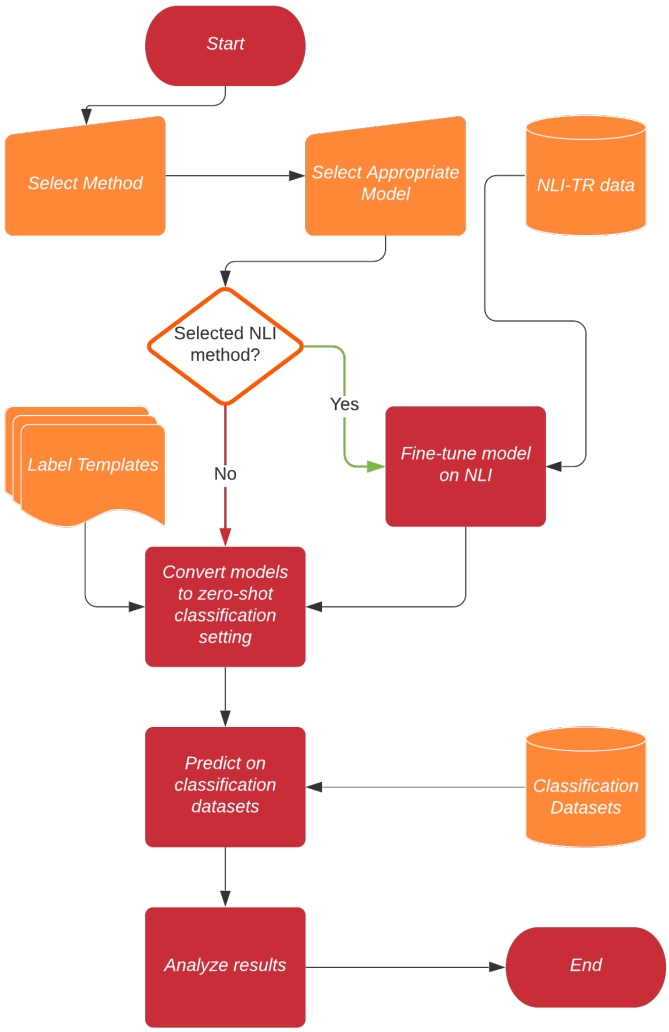


Figure 1: A general overview of the processes followed in the project.

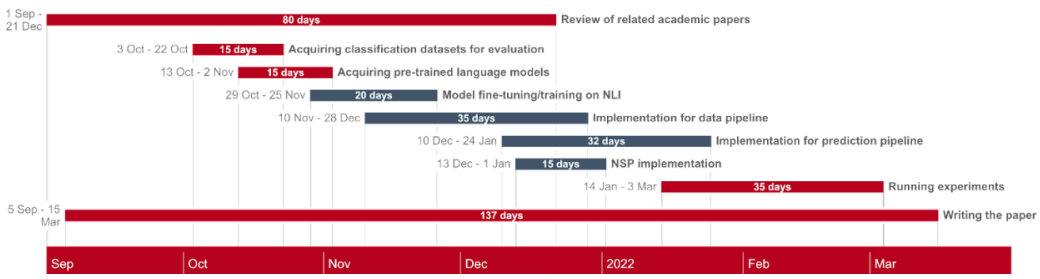


Figure 2: The project timeline.

4 Datasets

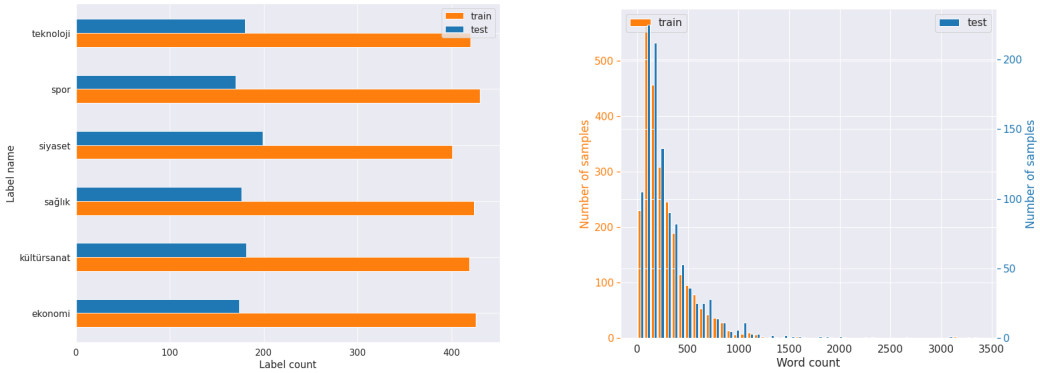
In this section, we present the unified dataset, which is a collection of 7 datasets widely used in other studies. In the following subsections we describe each dataset in detail. The subsections represent the three main categories that we assembled: topic, emotion and sentiment classification. Table 1 shows an overview of all the datasets with their corresponding label counts, sizes and categories.

Task	**Dataset**	**Label Count**	**Size**	**Test Size**
Topic Classification	TTC-3600	6	3600	0.3
	TTC-4900	7	4900	0.3
	TC-32	32	431306	0.3
Sentiment Classification	Tweet Sentiments	3	17289	3471
	Product Reviews	2	235165	0.3
Emotion Classification	TREMO	7	27350	0.3
	Blog Emotions	4	157	0.3

Table 1: General statistics of all classification datasets and train/test splits used in the study. Decimal values in the test size column represent the percentage of test samples, while integer values represent number of samples in the test set.

4.1 Topic Classification

TTC-3600 [20] created this dataset which consists of 3600 news articles from 6 different categories, and each category contains 600 articles. 2520 articles are used for training the supervised models while remaining 1080 is used for evaluation. Labels and their frequencies for train and test splits are given along with text length distributon in Figure 3. TTC-3600 is publicly available[4] and have been widely used in supervised text classification tasks by a great number of researchers [24, 25, 26]. Highest supervised classification accuracy was achieved by fine-tuning Turkish BERT model in [27] as 98.12%. Recently, [17] achieved 65.5% accuracy for zero-shot classification on TTC-3600, they used a multilingual version of RoBERTa which have been fine-tuned on NLI datasets.



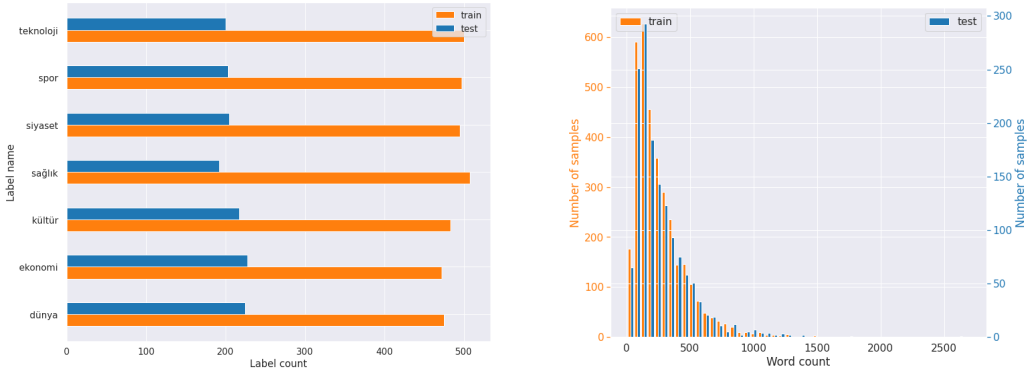
(a) (b)

Figure 3: (a) shows label frequencies for each split of the dataset TTC-3600 and (b) shows distribution of length of texts in the dataset in terms of word count.

TTC-4900 Similar to TTC-3600 this dataset contains 4900 news articles from 7 categories and each category has 700 examples. The dataset was made publicly available[5] by Kemik NLP Group[6]. TTC-4900 have also been used by researchers for supervised text classification [25, 28]. [29] achieved the highest accuracy score, 97.4%, on this dataset by fine-tuning Turkish DistilBERT for classification. Train and test splits

[4]https://github.com/denopas/TTC-3600
[5]https://www.kaggle.com/savasy/ttc4900
[6]http://www.kemik.yildiz.edu.tr/

consist of 3430 and 1470 news articles respectively, Figure 4 shows label frequencies in the dataset for each split.

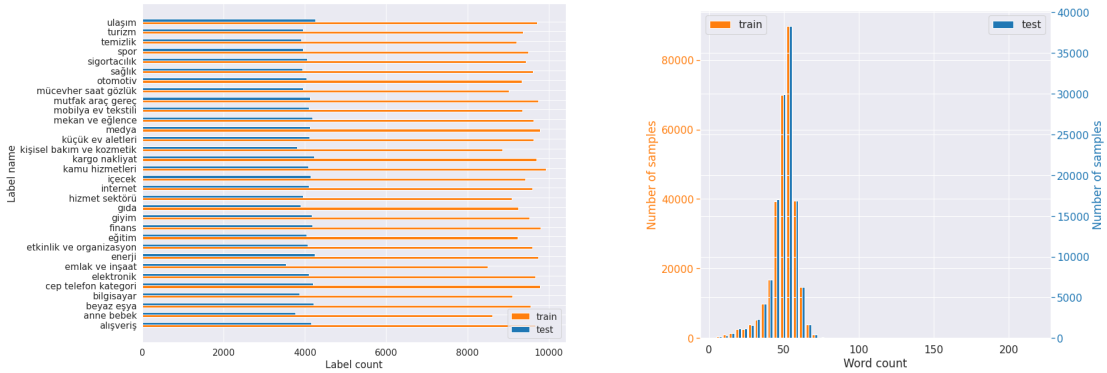


(a) (b)

Figure 4: (a) shows label frequencies for each split of the dataset TTC-4900 and (b) shows distribution of length of texts in the dataset in terms of word count.

TC-32 This dataset is contains user reviews for different domains, collected from various Turkish websites. There are 32 labels and 431,306 user reviews. We split the dataset into train and test sets which contain 301,914 and 129,392 reviews respectively. Label frequency distribution for each split is given in Figure 5. The dataset can be accessed from Kaggle[7].

[7]https://www.kaggle.com/savasy/multiclass-classification-data-for-turkish-tc32

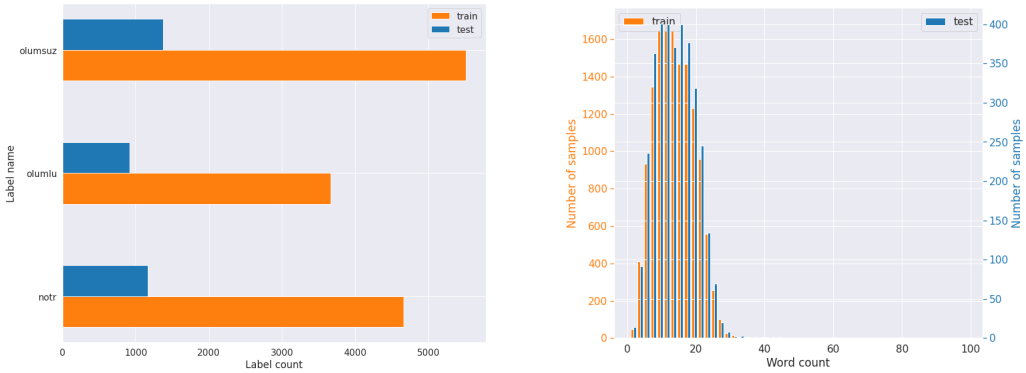


(a) (b)

Figure 5: (a) shows label frequencies for each split of the dataset TC-32 and (b) shows distribution of length of texts in the dataset in terms of word count.

4.2 Sentiment Classification

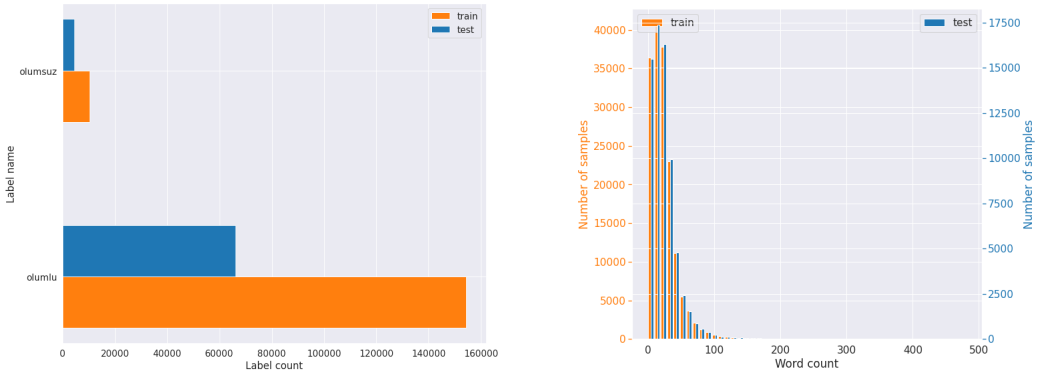
Tweet Sentiments We use the sentiment dataset that was presented in [30]. The dataset contains 17,289 Turkish tweets from 2011 to 2017 about a GSM operator. [31] applied wide variety of text representation and classification methods on the dataset and reported the highest accuracy score achieved as 82.14%. Train and test sets contain 13,832 and 3,457 tweets respectively.



(a) (b)

Figure 6: (a) shows label frequencies for each split of the dataset Tweet Sentiments and (b) shows distribution of length of texts in the dataset in terms of word count.

Product Reviews Product reviews dataset consists of online reviews collected from various websites. Reviews are separated into two categories, negative and positive. The dataset is available on HuggingFace Hub.[8]



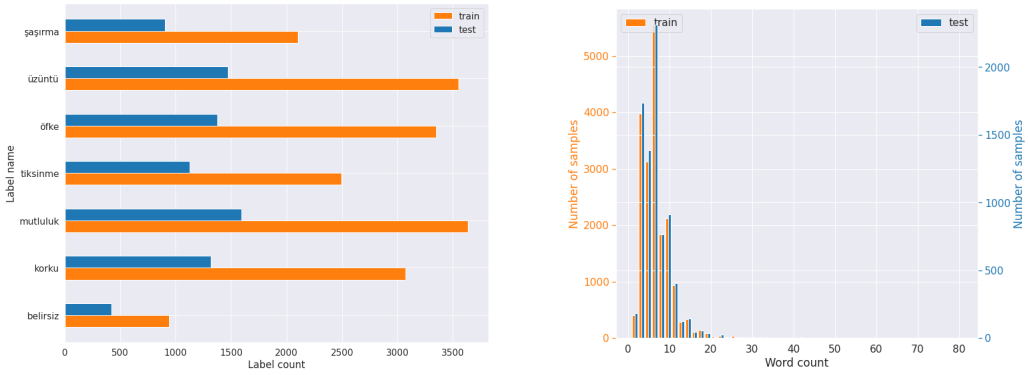
(a) (b)

Figure 7: (a) shows label frequencies for each split of the dataset Product Reviews and (b) shows distribution of length of texts in the dataset in terms of word count.

[8]https://huggingface.co/datasets/turkish_product_reviews

4.3 Emotion Classification

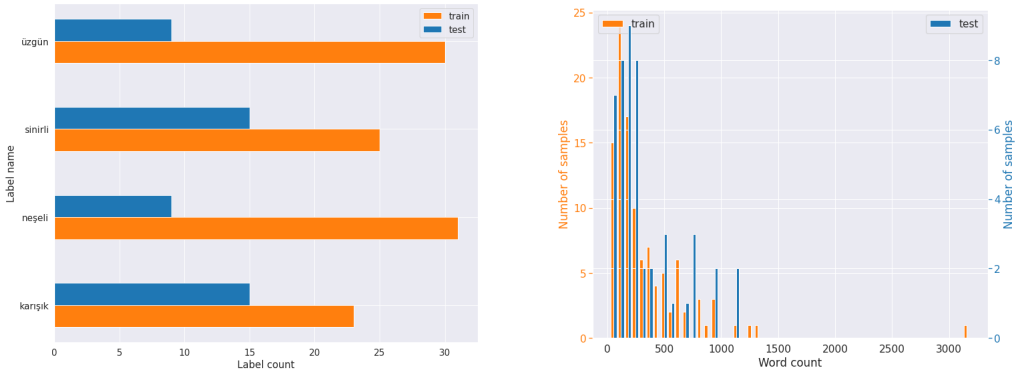
TREMO TREMO [32] is a manually annotated collection of memories and experiences of 5000 people from different ages and different living areas. The people participated in the study were asked to share their memories that fit into six categories of emotions. After the survey, data was validated by a team of annotators. Forementioned emotion categories are happiness, fear, anger, sadness, disgust and surprise. There is also an extra category for ambiguous texts. Distribution of these categories can be seen in Figure 8. [27] achieved the highest supervised classification accuracy, 93.63%, on this dataset by fine-tuning BERT.



(a) (b)

Figure 8: (a) shows label frequencies for each split of the dataset TREMO and (b) shows distribution of length of texts in the dataset in terms of word count.

Blog Emotions Blog Emotions dataset [33] consists of manually annotated 157 blog posts collected from different websites. Blog posts are categorized into 4 classes, happy, angry, sad and mixed. Figure 9 shows the distribution of posts for the corresponding classes. [33] achieved an accuracy score of 85.23% on this dataset by using character level n-grams and binary features.



(a) (b)

Figure 9: (a) shows label frequencies for each split of the dataset Blog Emotions and (b) shows distribution of length of texts in the dataset in terms of word count.

5 Methodology

All methods that are used in this study to implement zero-shot text classification utilize label meanings in order to make predictions without training on a specified classification task. Methods utilize label meanings by following a common idea: the use of templates. This idea will be further explained in the following inference sections.

Although the listed methods require a training stage, for NSP and MLM approaches this training stage is already done in the language model pre-training. However, NLI method requires fine-tuning on a specific dataset before making any zero-shot predictions. A general overview of the zero-shot classification process is presented in Figure 1.

5.1 Natural Language Inference

NLI is a text pair classification task. Given two input texts, namely premise and hypothesis, the aim is to determine if relation between these texts is entailment, contradiction or neutral.

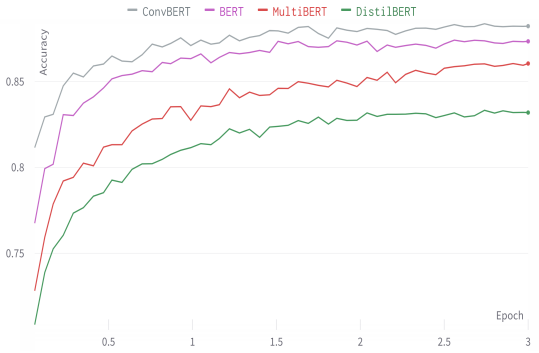
5.1.1 Training Stage

We use NLI-TR dataset [21] to fine-tune the models and we access it from Hugging Face Hub[9]. The dataset consists of machine translated Turkish versions of SNLI [22] and MultiNLI [23] datasets. We train 4 models (ConvBERT[10], BERT[11], DistilBERT[12], MultiBERT[13]) with the same set of hyperparameters for 3 epochs. A list of the fine-tuned models and accuracy scores are given in Table 2 and additional training details are publicly available[14]. We use the same hyperparameters for each model. Models are trained with Adam optimizer for 3 epochs and we start the learning rate at the value 2e-05 and use linear scheduling, with batch size of 32.

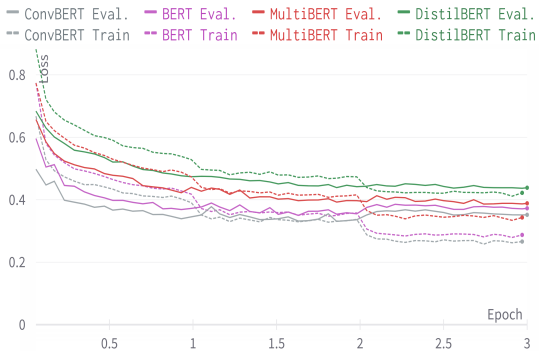
	MultiNLI-TR		**SNLI-TR**		**AllNLI-TR**	
	Mismatched Dev	**Matched Dev**	**SNLI Test**	**SNLI Dev**	**Mismatched Dev**	**SNLI Test**
BERT	0.80	0.794	0.872	0.873	0.799	0.874
DistilBERT	0.736	0.728	0.838	0.832	0.738	0.839
ConvBERT	**0.818**	**0.815**	0.88	**0.882**	0.812	**0.888**
MultiBERT	0.768	0.757	0.857	0.8605	0.765	0.856

Table 2: Accuracy results of the fine-tuned transformer models on NLI datasets. AllNLI-TR is merged and shuffled version of the MultiNLI-TR and SNLI-TR datasets. Highest accuracies achieved in each test and development set are given in bold.

[9] https://huggingface.co/datasets/nli_tr
[10] https://huggingface.co/dbmdz/convbert-base-turkish-mc4-cased
[11] https://huggingface.co/dbmdz/bert-base-turkish-cased
[12] https://huggingface.co/dbmdz/distilbert-base-turkish-cased
[13] https://huggingface.co/bert-base-multilingual-cased
[14] https://wandb.ai/emrecncelik/zeroshot-turkish?workspace=user-emrecncelik

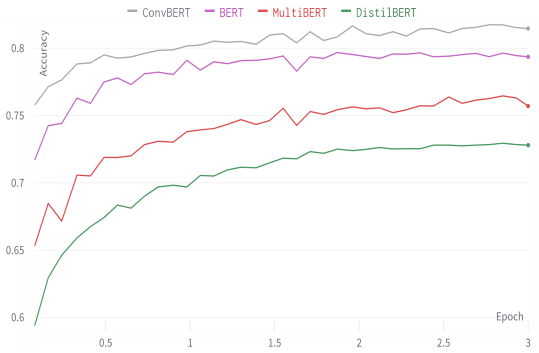


(a) Fine-tuning accuracy results on SNLI-TR

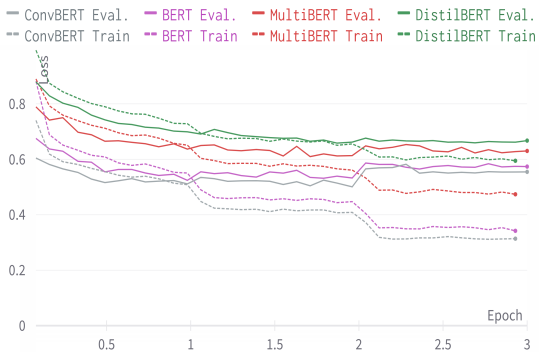


(b) Fine-tuning loss results on SNLI-TR

Figure 10: (a) and (b) show model accuracies and losses recorded during fine-tuning on the SNLI-TR dataset. Accuracy is calculated on development split of the dataset, while loss is calculated on both development and train splits.

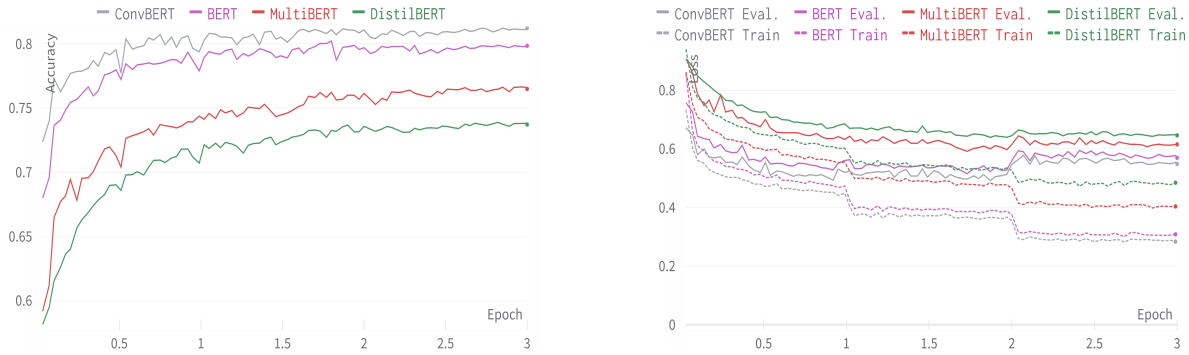


(a) Fine-tuning accuracy results on MultiNLI-TR



(b) Fine-tuning loss results on MultiNLI-TR

Figure 11: (a) and (b) show model accuracies and losses recorded during fine-tuning on the MultiNLI-TR dataset. Accuracy is calculated on matched development split of the dataset, while loss is calculated on both matched development and train splits.



(a) Fine-tuning accuracy results on AllNLI-TR

(b) Fine-tuning loss results on AllNLI-TR

Figure 12: (a) and (b) show model accuracy and losses during recorded fine-tuning on the AllNLI-TR dataset. Accuracy is calculated on matched development split of MultiNLI-TR dataset, while loss is calculated on matched development split of MultiNLI-TR and train split of AllNLI-TR.

5.1.2 Inference Stage

In order to utilize NLI models for zero-shot classification, we follow a similar approach to the one presented in [13], and use the HuggingFace [34] pipeline. For a dataset with j number of labels, we consider the input text, x_i, as the premise and the template, t_j, as the hypothesis. For each template generated from the labels, model makes an entailment prediction and the template with the maximum entailment score is considered as the class of the input text, Figure 13 explains this idea.

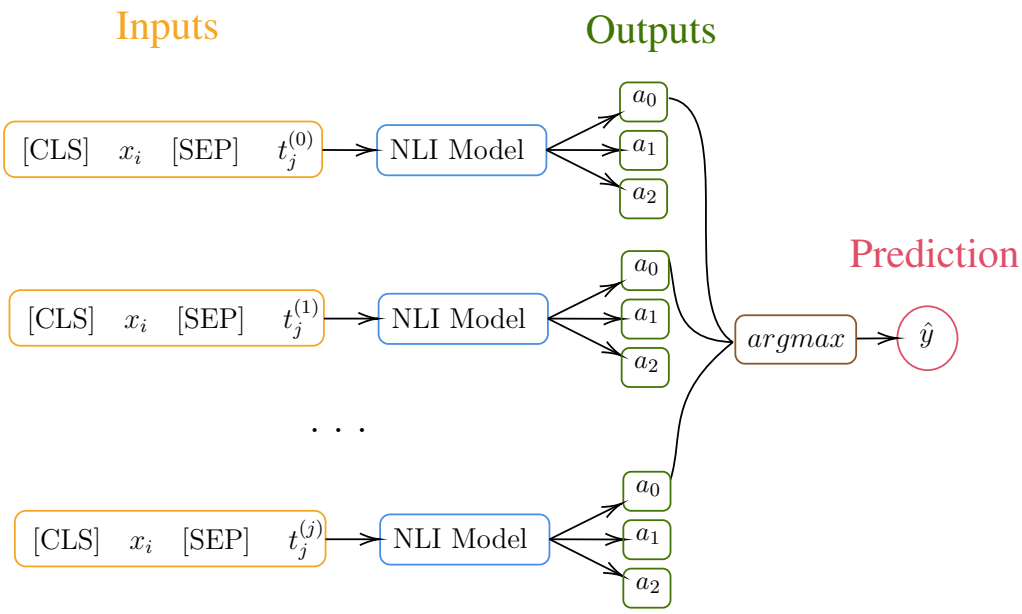


Figure 13: NLI model is applied to zero-shot classification as shown in this diagram. x_i, t_j, a and $\hat{y}$ represent input text, template, output of the last layer and the predicted class respectively. a_0 is considered to be the entailment output. Outputs for the classes contradiction and neutral are not used. [CLS] and [SEP] are special tokens used to separate sentences.

5.2 Next Sentence Prediction

NSP is a binary text-pair classification task that is used in pre-training BERT, and aims to incorporate understanding of sentence relationships into the language model. This pre-training task is unique to BERT architecture, and was abandoned in language models with similar architectures (eg. RoBERTa).

5.2.1 Training Stage

As explained in [14], during pre-training the model tries to predict if sentence `A` and `B` are consecutive sentences. 50% of the time sentence `B` actually follows sentence `A`, and in the other %50 sentence `B` is a random sentence sampled from the training corpus. Since the model was trained previously by other researchers, there is no need for further fine-tuning[15] to perform zero-shot classification like the NLI setting.

[15] Although not required, might be a viable solution to few-shot classification with NSP.

5.2.2 Inference Stage

The idea we follow to apply NSP to zero-shot classification [16] is very similar to the one in the NLI section. Given the input text, x_i, and label templates, t_j, we consider x_i and t_j as sentence `A` and sentence `B` and try to predict if x_i and t_j are consecutive. We take the template with the maximum of positive output as the class prediction of the model, Figure 14 explains this idea.

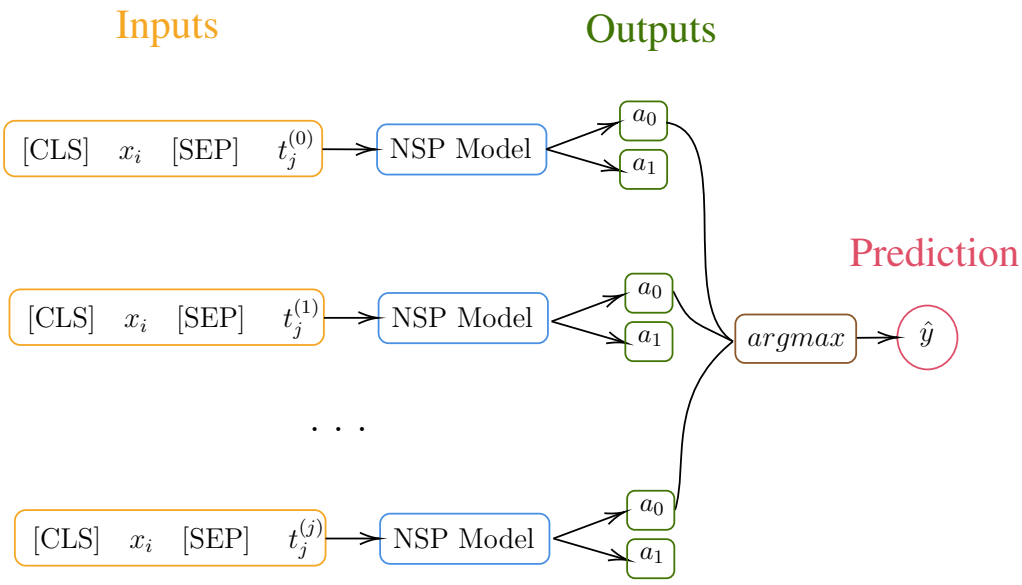


Figure 14: NSP model is applied to zero-shot classification as shown in this diagram. x_i, t_j, a and $\hat{y}$ represent input text, template, output of the last layer and the predicted class respectively. a_0 is considered to be the isNext output. Outputs for the class notNext is not used. [CLS] and [SEP] are special tokens used to separate sentences.

5.3 Masked Language Modeling with Word Embeddings

Masked Language Modeling is a popular pre-training task amongst the models with the transformer architecture, which include BERT. In MLM, the language model tries to predict a token depending on the surrounding context tokens. Along MLM, we also use a word embedding model, FastText [35], to find similarities of predicted candidate tokens with the labels. FastText is a skipgram based word embedding model, which represents words as character level n-grams in order to be able to handle out-of-vocabulary words, unlike other word embedding techniques such as Word2Vec [9].

5.3.1 Training Stage

Similar to NSP, training for the MLM is done at the pre-training stage therefore we do not apply any additional training. In the pre-training stage 15% of the token positions are randomly selected from the corpus, and are replaced with a mask token with 80% chance, in this case: `[MASK]`. 10% of the time selected token remains the same, and in the other 10% selected token is replaced with a random token. Then the other tokens in the input sequence are used to predict the original token. FastText is also pre-trained on the Turkish part of the Wikipedia and we do not any further training.

5.3.2 Inference Stage

Similar to other methods we have shown, we use templates. However, this time replace the {} token in the with the `[MASK]` token, instead of replacing the with the candidate labels. After this replacement, we append the template at the end of the text to be classified. MLM model predicts the top 5 most suitable candidate tokens for the mask and we take an average vector of these candidates using the word embedding model. Finally we extract the word vectors of each candidate label, and calculate the cosine similarity of labels to the average vector, most similar label is selected to be the predicted class, as shown in the Figure 15.

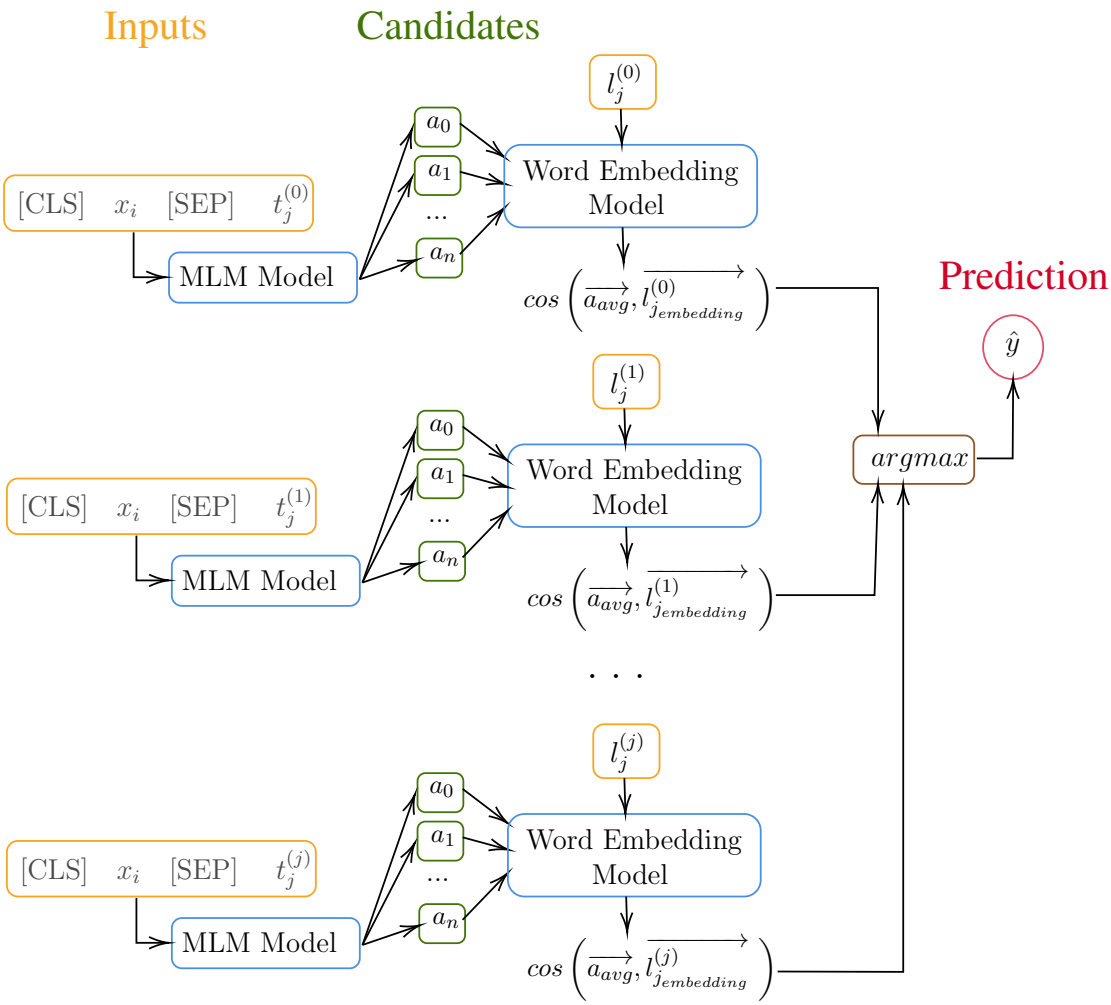


Figure 15: MLM model is applied to zero-shot classification as shown in this diagram. x_i, t_j, a_n, l_j and $\hat{y}$ represent input text, template, candidate tokens, candidate label and the predicted class respectively. Values a_0 to a_j are considered to be the candidate tokens. [CLS] and [SEP] are special tokens used to separate sentences.

6 Experimental Setup

We run the experiments for every combination of each dataset, template, model, and zero-shot method, so we run 168 experiments in total.

6.1 Templates

The templates mentioned in previous sections are generated manually and list of generated templates can be seen in the Table 3. We use different templates for each context, since templates which are related with the input text, and closer to natural language give better results as shown in previous studies [13, 17, 16].

Context	Template
Emotion, Sentiment	T_{E_0}: Bu metnin içerdiği duygu çoğunlukla {} T_{E_1}: Bu metin çoğunlukla {} duygular içeriyor T_{E_2}: Çoğunlukla {} duygular hissediyorum
News	T_{N_0}: Bu haberin konusu çoğunlukla {} T_{N_1}: Bu haber çoğunlukla {} ile ilgilidir T_{N_2}: Bu haberin içeriği çoğunlukla {} ile ilgilidir
Review	T_{R_0}: Bu şikayetin konusu çoğunlukla {} T_{R_1}: {} hizmetinizden memnun değilim T_{R_2}: Bu şikayetin içeriği çoğunlukla {} ile ilgili

Table 3: List of all templates used in experiments. {} symbols are replaced with labels during inference.

6.2 Models

All models we use for corresponding methods are presented in Table 4. Models used for NSP are limited to BERT models that support NSP since NSP is a unique pre-training task of BERT. NLI models are the models we have fine-tuned, we show their training results in the Methodology section in Table 2. We use same models for MLM as NSP, and additionally a fastText model. Checkpoint we used for the fastText model was trained on the Turkish Wikipedia and can be accessed on the official fastText website[16].

[16]https://fasttext.cc/docs/en/pretrained-vectors.html

Method	Model Checkpoint
NLI	emrecan/distilbert-base-turkish-cased-allnli_tr emrecan/bert-base-turkish-cased-allnli_tr emrecan/convbert-base-turkish-mc4-cased-allnli_tr emrecan/bert-base-multilingual-cased-allnli_tr
NSP	dbmdz/bert-base-turkish-cased dbmdz/bert-base-turkish-128k-cased
MLM	dbmdz/bert-base-turkish-cased dbmdz/bert-base-turkish-128k-cased

Table 4: List of model checkpoints for each method.

7 Results

7.1 Experiments

We observe that $\text{ConvBERT}_{\text{NLI}}$ generally outperforms BERT and other variants on most datasets due to its bigger corpus, Turkish part of mC4, [36] and due to the architectural improvements, which primarily consist of replacing the redundant attention heads with span-based dynamic convolution as presented in [37]. Table 5, also shows that when we use same BERT models, NLI method yields the best scores on all datasets except TREMO, in which NSP performs better than NLI. MLM with word embeddings method only performs better in TTC-4900 when compared with other methods that use the same BERT model. Experiment results also show that MLM works better with templates that have the mask token somewhere at the middle of the template rather than at the start or the end. Results of all these experiments are also available in our Wandb project[17].

[17]https://wandb.ai/emrecncelik/zeroshot-turkish-predictions?workspace=user-emrecncelik

Dataset	Template	DistilBERT$_{NLI}$	BERT$_{NLI}$	ConvBERT$_{NLI}$	mBERT$_{NLI}$	BERT$_{NSP}$	BERT128k$_{NSP}$	BERT$_{MLM}$	BERT128k$_{MLM}$
TTC-3600	T_{N_0}	0.33	0.54	**0.73**	0.56	0.45	0.62	0.07	0.20
	T_{N_1}	0.34	0.63	**0.79**	0.59	0.51	0.63	0.43	0.45
	T_{N_2}	0.34	0.57	**0.78**	0.6	0.56	0.67	0.49	0.45
TTC-4900	T_{N_0}	0.23	0.18	**0.62**	0.44	0.44	0.6	0.15	0.19
	T_{N_1}	0.27	0.23	**0.66**	0.46	0.32	0.55	0.41	0.37
	T_{N_2}	0.26	0.2	**0.64**	0.49	0.35	0.53	0.42	0.38
Tweet Sentiments	T_{E_0}	0.48	0.42	**0.56**	0.48	0.37	0.38	0.23	0.28
	T_{E_1}	0.47	0.43	**0.55**	0.4	0.28	0.21	0.30	0.33
	T_{E_2}	0.45	0.5	**0.58**	0.41	0.33	0.41	0.31	0.31
TREMO	T_{E_0}	0.47	0.35	0.51	0.44	0.52	**0.61**	0.07	0.20
	T_{E_1}	0.35	0.26	0.47	0.47	0.55	**0.6**	0.11	0.13
	T_{E_2}	0.38	0.51	0.47	0.35	**0.64**	0.63	0.05	0.05
Blog Emotions	T_{E_0}	0.35	**0.4**	0.33	0.39	0.28	0.18	0.05	0.20
	T_{E_1}	0.38	**0.5**	0.39	0.37	0.26	0.33	0.28	0.16
	T_{E_2}	0.44	**0.48**	0.3	0.32	0.31	0.37	0.26	0.18
TC-32	T_{R_0}	0.13	0.25	0.26	0.07	0.11	**0.36**	0.04	0.06
	T_{R_1}	0.07	0.27	0.23	0.07	0.11	**0.36**	0.03	0.06
	T_{R_2}	0.18	0.22	0.26	0.12	0.12	**0.40**	0.25	0.28
Product Reviews	T_{E_0}	0.84	**0.92**	**0.92**	0.85	0.37	0.39	0.32	0.10
	T_{E_1}	0.87	0.92	**0.93**	0.85	0.16	0.07	0.15	0.20
	T_{E_2}	0.76	**0.92**	0.91	0.77	0.36	0.63	0.32	0.42

Table 5: Weighted average F1 scores of all models evaluated on test sets of all datasets. We use the three different templates for each context which were presented in Table 3. Underlined values represent the best scores of the methods with the same BERT models. Bold values represent the best scores across all model and method combinations.

Comparison in the Figure 16 shows us while NLI is generally the best choice, NSP can outperform NSP in some emotion classification tasks. MLM with word embeddings yields scores that are worse than other methods in all cases. Also, we see that NLI significantly performs better than others in sentiment classification tasks.

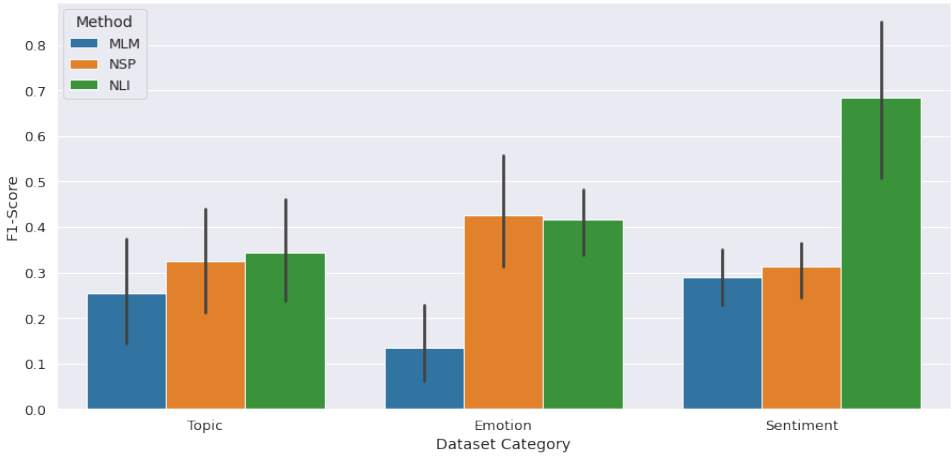


Figure 16: All methods are compared on the three different categories of the datasets, black line represents the standard deviation and the bars represent mean of the weighted average f1-score of the method on a specified category. This comparison is only done for the BERT model, since that is the only common model between the methods.

We also compare all our models with the previous studies in zero-shot Turkish text classification[17], as seen in Figure 6. This comparison is only done on TTC-3600 dataset since it is the only common dataset we were able to access. Even though XLM-RoBERTa [38, 18] is a more optimized, multilingual version of the BERT, trained with a larger multilingual corpora, monolingual BERT_{NLI} with a smaller corpus performs only 3% worse than the XLM-RoBERTa. Additionally, $\text{BERT128k}_{\text{NSP}}$ outperforms XLM-RoBERTa without any prior training on the NLI dataset and $\text{ConvBERT}_{\text{NLI}}$ model performs drastically better than XLM-RoBERTa on this particular dataset.

	$\text{DistilBERT}_{\text{NLI}}$	**BERT_{NLI}**	**$\text{ConvBERT}_{\text{NLI}}$**	**$\text{mBERT}_{\text{NLI}}$**	**BERT_{NSP}**	**$\text{BERT128k}_{\text{NSP}}$**	**BERT_{MLM}**	**$\text{BERT128k}_{\text{MLM}}$**	**XLM-RoBERTa (Birim et al., 2021)**
Accuracy	0.36	0.64	**0.79**	0.62	0.55	0.69	0.49	0.49	0.66

Table 6: Comparison of accuracy of the models on the TTC-3600 dataset.

7.2 Demo Application

We release a demo application to share all trained models in one place, and allow readers to experiment themselves with different methods and models. The application was implemented using Streamlit[18] and it is hosted on HuggingFace Spaces.

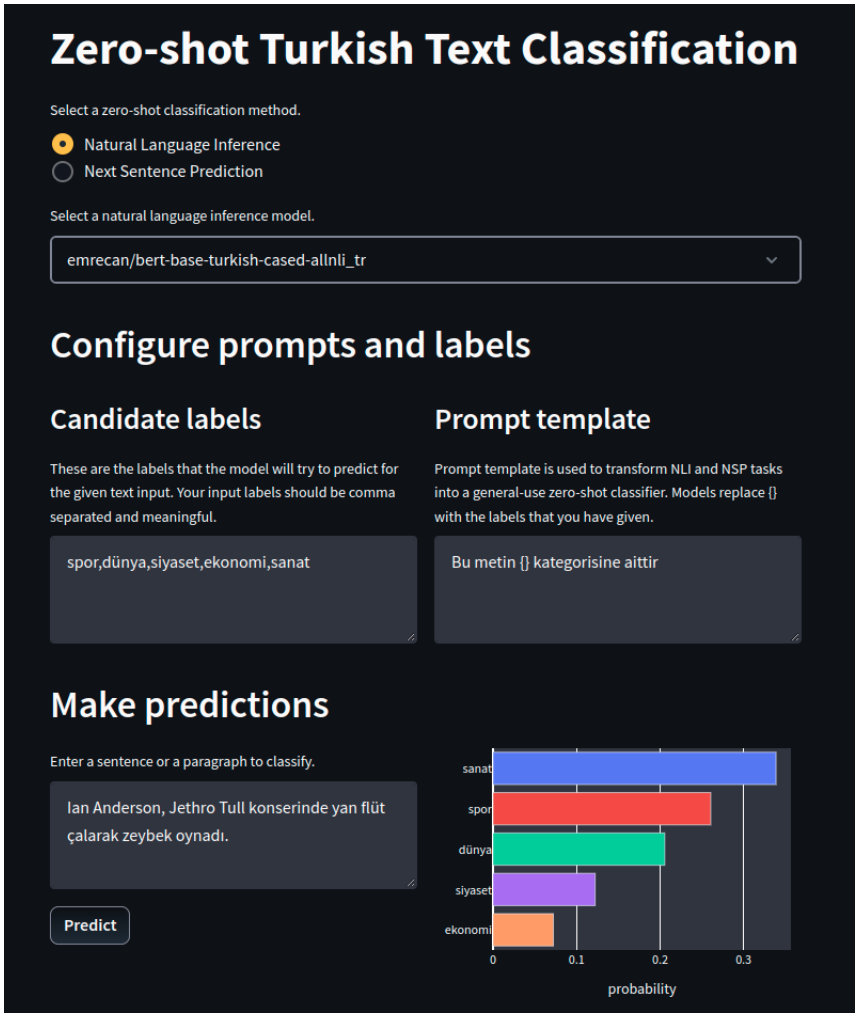


Figure 17: Screenshot of the demo application from HuggingFace Spaces

[18]https://github.com/streamlit/streamlit

8 Conclusions and Future Work

Our analysis of the experimental results show that NLI stands to be the best choice for most datasets amongst the methods we have presented. However, not all languages have an NLI dataset for the training, in such cases NSP seems to be a good alternative for NLI. Additionally, we see that use of monolingual models yield similar or better scores than the multilingual models on zero-shot classification, as expected. Finally, our results suggest that MLM with word embeddings method is not a viable solution to the zero-shot text classification problem, due to its inferior performance in almost all settings.

In the future, for approaches that do not require any labeled data to perform zero-shot classification we may improve the zero-shot performance of such methods by continuing the pre-training process on a set of unlabeled data related to the final task. Furthermore, such methods may also be applied to few-shot classification, since we can generate a bigger sample from a few examples by using a diverse set of templates.

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